



Fig. 2: Thermal imaging at work in a retail outlet

it, enabling quicker billing as the cart can communicate with the cash register. The customer's purchase history is also maintained, which enables the cart to anticipate which direction the customer is likely to move in through the retail shop floor. Customised suggestions then begin to pop up on the screen, to help the shopper.

Thermal imaging. Thermal imaging cameras detect radiation in the infrared range of the electromagnetic spectrum (roughly 9000-14,000 nanometres or 9-14 μm) and generate images of that radiation, known as thermograms. All objects emit infrared radiation, which is above absolute zero, and the amount of radiation emitted by an object increases with temperature. Since the temperature of human beings is higher than most surrounding objects', thermography allows detection and counting of humans. Array sensors detect heat sources in thermal imaging systems.

These systems usually use embedded technology and are placed at a height for precise results. As they detect the heat emitted from people, they are able to count well even in different intensities of lighting, and need not employ complex background removal algorithms unlike computer vision systems. This results in a more precise people count. However, the system may fail in cold weather when customers are wearing thick woollens.

People counters help collect information such as customer footfalls and the path taken by the customers in the store, helping to generate customer insights such as their reactions to in-store promotions and price changes, and linkages between footfalls and sales. This information ultimately helps in improving the store layout and quality of service.

SixthSense technology. Many of you would have seen that amazing video on TED, featuring Pranav Mistry's SixthSense technology. It is in development stage at MIT Media Lab's Fluid Interfaces Group and is "a wearable gestural interface that augments the physical world around us with digital information." This technology will make the product shelves in retail malls come alive by providing interaction with product packages.

Generally, when shopping for a high-value product such as a laptop or a car, we are likely to refer to price and feature comparison websites. However, small ticket items such as toys and music CDs are more of impulse purchases and we usually do not have time to read product reviews before making the purchase.

SixthSense technology enables the customer to gain access to a whole lot of product-related information, such as reviews and comparisons by merely scanning the bar code of the product.

The entire information can be projected on any surface. This technology could also give stores the liberty to stock only one sample of each product and offer delivery the next day after the customer has selected the product and purchased it by scanning the barcode using his mobile.

Augmented reality. Augmented reality technology is becoming increasingly popular. It adds a visual layer of information on top of surfaces such as a mirror and the content interacts with layers of information provided by the real world. For example, by

holding up a specially coded box or label to a webcam-equipped computer, the user can see an augmented, 3D image of the product, or a graphic display of other additional information.

Intel has developed a high-technology mirror that lets shoppers see how the chosen clothes would look on them. It is a digital trial room where the customer just needs to stand in front of an LCD monitor and parametric technology simulates his body type displaying how chosen garments would fit on him—based on weight, height and measurements.

Malls of the future could use digital signage that delivers targeted advertising and also captures key customer demographic information via video for analysis later, thus creating almost real-time customer feedback.

Major consumer product companies are also testing the business value of mobile mash-ups. Bionic Eye is an iPhone app wherein the user can point the camera at a mall to get a screen display of all the outlets inside the building. Virtual signposts with directions to nearby points of interest are also included. Interestingly, all this works offline.

Data mining and predictive analytics

'Big Data'—a term much in vogue today—refers to deployment of huge